

Dominick Reilly

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Education

University of North Carolina at Charlotte

Aug 2022 - Present

Ph.D. Student in Computer Science

Advisor: Dr. Srijan Das

University of North Carolina at Charlotte

Jan 2019 - May 2022

Bachelor of Science in Computer Science, GPA: 4.0

Experience

University of North Carolina at Charlotte

Aug 2022 - Present

Research Assistant - Charlotte Machine Learning Lab

- Domain adaption of Vision Language Models (VLMs) through visual probing, enabling adaptation to novel visual domains and preventing catastrophic forgetting of source domains. [1]
- Cross-view (egocentric → exocentric) knowledge transfer in VLMs for enhancing egocentric understanding in exocentric videos. [6, 2]
- Multi-modal (RGB + Human Pose + Human Object Interactions) Large Language Models (LLM) for understanding daily living actions from video. Our approach leverages a curriculum-based training strategy for integrating new modalities into the LLM embedding space. [4]
- Multi-modal (RGB + Human Pose) and viewpoint agnostic Vision Transformers (ViT) for understanding subtle appearance and motion cues in human activity videos. [10]
- Self-supervised learning and masked autoencoders for training ViTs on small, out-of-domain data distributions. [11]

Honda Research Institute, California

Jun 2024 - Sep 2024

Student Researcher - Perception Team

- View-invariant VLMs for procedural video understanding. View-invariant representations are learned on cross-view videos (ego & exo) and improve performance on single-view videos.
- Learning pose-augmented RGB representations in CNNs for weakly-supervised action and error detection in procedural activity videos.

Inria, Sophia Antipolis, France

Jan 2024 - Jun 2024

Research Intern - STARS Team (Chateaubriand Fellow)

- Multi-modal (RGB + Human Pose) CLIP models for improved zero-shot understanding of daily living activities from video. [8] [5]
- Learnable gating mechanisms to improve feature selection for temporal activity detection.[9]

University of North Carolina at Charlotte

Jul 2021 - Jul 2022

Research Assistant - Data Privacy Lab

- Safeguarding face and iris images from deep-learning based re-identification attacks. [13]
- Created interactive webpage demonstrating safeguards on face and iris images. Try the demo for yourself at <http://3.223.148.187/>. [12]

Publications (C)onference, (P)reprint, (W)orkshop

1. (P) **Dominick Reilly**, Manish Kumar Govind, Le Xue, Srijan Das, “VisCoP: Visual Probing for Video Domain Adaptation of Vision Language Models,” [arXiv:2510.13808](#), 2025.
2. (W) **Dominick Reilly**, Manish Kumar Govind, Le Xue, Srijan Das, “From My View to Yours: Ego-Augmented Learning in Large Vision Language Models for Understanding Exocentric Daily Living Activities,” MMFM3: The 3rd Workshop on What is Next in Multimodal Foundation Models? at [CVPR](#), 2025.
3. (W) **Dominick Reilly**, Srijita Das, Srijan Das, “Learning Pose-aware Representations in Vision Transformers for Understanding Activities of Daily Living,” 8th Workshop and Competition on Affective & Behavior Analysis in-the-wild at [CVPR](#), 2025.
4. (C) **Dominick Reilly**, Rajat Subhra Chakraborty, Arkaprava Sinha, Manish Kumar Govind, Pu Wang, Francois Bremond, Le Xue, Srijan Das, “LLAVIDAL: A Large Language Vision Model for Daily Activities of Living,” IEEE/CVF Conference on Computer Vision and Pattern Recognition ([CVPR](#)), 2025.
5. (C) Arkaprava Sinha, **Dominick Reilly**, Francois Bremond, Pu Wang, Srijan Das, “SKI Models: Skeleton Induced Vision-Language Embeddings for Understanding Activities of Daily Living,” Proceedings of the AAAI Conference on Artificial Intelligence ([AAAI](#)), 2025.
6. (P) **Dominick Reilly**, Manish Kumar Govind, Le Xue, Srijan Das, “From My View to Yours: Ego-Augmented Learning in Large Vision Language Models for Understanding Exocentric Daily Living Activities,” [arXiv:2501.05711](#), 2024.
7. (P) Ali Khaleghi Rahimian, Manish Kumar Govind, Subhajit Maity, **Dominick Reilly**, Christian Kümmerle, Srijan Das, Aritra Dutta, “Fibottention: Inceptive Visual Representation Learning with Diverse Attention Across Heads,” [arXiv:2406.19391](#), 2024.
8. (W) Mahmoud Ali, Di Yang, Arkaprava Sinha, **Dominick Reilly**, Srijan Das, Gianpiero Francesca, Francois Bremond, “Quo Vadis, Video Understanding with Vision-Language Foundation Models?,” Workshop on Video-Language Models ([NeurIPS](#)), 2024.
9. (W) Aglind Reka, Diana Laura Borza, **Dominick Reilly**, Michal Balazsia, Francois Bremond, “Introducing Gating and Context into Temporal Action Detection,” Workshop on Affective Behavior Analysis in-the-wild at ([ECCV](#)), 2024.
10. (C) **Dominick Reilly** and Srijan Das, “Just Add π ! Pose Induced Video Transformers for Understanding Activities of Daily Living,” IEEE/CVF Conference on Computer Vision and Pattern Recognition ([CVPR](#)), 2024.
11. (C) Srijan Das, Tanmay Jain, **Dominick Reilly**, Pranav Balaji, Soumyajit Karmakar, Shyam Marjit, Xiang Li, Abhijit Das, Michael Ryoo, “Limited Data, Unlimited Potential: A Study on ViTs Augmented by Masked Autoencoders,” IEEE/CVF Winter Conference on Applications of Computer Vision ([WACV](#)), 2024.
12. (C) Muhammad Saleem, **Dominick Reilly**, Liyue Fan, “DP-Shield: Face Obfuscation with Differential Privacy,” International Conference on Extending Database Technology, 2022.
13. (C) **Dominick Reilly**, Liyue Fan, “Comparative Evaluation for Differentially Private Image Obfuscation,” IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications ([IEEE TPS](#)), 2021.

Patents filed

1. Srijan Das and **Dominick Reilly**. Large Language Vision Model (LLVM), US Patent Application No. 63/693,982 filed on Sep 12, 2024.

Academic Activities

- Reviewer for CVPR 2025, AAAI 2023/24/25

Awards & Achievements

1. Our research “Computer model aims to enhance video technology” highlighting the implications of AI video analysis is aired on WSOC-TV ([Link](#))
2. CVPR 2025 Travel Award
3. The Chateaubriand Fellowship (awarded by the Embassy of France), 2023
4. Best poster award in Mathematics and Computer Science, UNC Charlotte Undergraduate Research Conference, 2020